Tier 1 Tier 2 Tier 3 Tier 4

(a) r = 0.02
eSEN-30M-OAM
GRACE-2L-OAM
EquiformerV2
MACE-MP-0
UMA-M-P1
UMA-S-P1
MatterSim-v1.0.0-5M
orb-v2
orb-v3-direct-20-omat
orb-v3-conservative-inf-omat
NequIP-OAM-L
MACE-MPA-0
PET-OMAT-XL
MACE-MH-1-OMAT
PET-OAM-XL

(b) r = -0.09
eSEN-30M-OAM
GRACE-2L-OAM
EquiformerV2
MACE-MP-0
MatterSim-v1.0.0-5M
orb-v2
NequIP-OAM-L
MACE-MPA-0
orb-v3-conservative-inf-omat
PET-OAM-XL

(c) r = 0.18
eSEN-30M-OAM
EquiformerV2
GRACE-2L-OAM
MACE-MP-0
NequIP-OAM-L
orb-v2
MatterSim-v1.0.0-5M
MACE-MPA-0
orb-v3-conservative-inf-omat
PET-OAM-XL

RDF error (without hydrogen) [%]

(d) r = 0.65
MACE-MPA-0
MACE-MP-0
orb-v3-direct-20-omat
MatterSim-v1.0.0-5M
EquiformerV2
orb-v3-conservative-inf-omat
NequIP-OAM-L
orb-v2
PET-OAM-XL
GRACE-2L-OAM
PET-OMAT-XL
eSEN-30M-OAM
UMA-S-P1
MACE-MH-1-OMAT
UMA-M-P1

(e) r = -0.68
MACE-MP-0
MACE-MPA-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v3-conservative-inf-omat
orb-v2
NequIP-OAM-L
PET-OAM-XL
GRACE-2L-OAM
eSEN-30M-OAM

(f) r = 0.27
MACE-MPA-0
MACE-MP-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v3-conservative-inf-omat
orb-v2
NequIP-OAM-L
PET-OAM-XL
GRACE-2L-OAM
eSEN-30M-OAM

Pressure histogram error (without hydrogen) [%]

(g) r = 0.56
MACE-MP-0
GRACE-2L-OAM
eSEN-30M-OAM
EquiformerV2
orb-v2
NequIP-OAM-L
MACE-MPA-0
MatterSim-v1.0.0-5M
MACE-MH-1-OMAT
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
UMA-M-P1
UMA-S-P1
PET-OAM-XL
PET-OMAT-XL

(h) r = -0.57
MACE-MP-0
eSEN-30M-OAM
EquiformerV2
GRACE-2L-OAM
orb-v2
NequIP-OAM-L
MatterSim-v1.0.0-5M
MACE-MPA-0
orb-v3-conservative-inf-omat
PET-OAM-XL

(i) r = 0.31
MACE-MP-0
GRACE-2L-OAM
EquiformerV2
eSEN-30M-OAM
orb-v2
NequIP-OAM-L
MatterSim-v1.0.0-5M
MACE-MPA-0
orb-v3-conservative-inf-omat
PET-OAM-XL

VDOS error (without hydrogen) [%]

Force RMSE (without hydrogen) [eV/Å]

F1 score

κ_{SRME} score